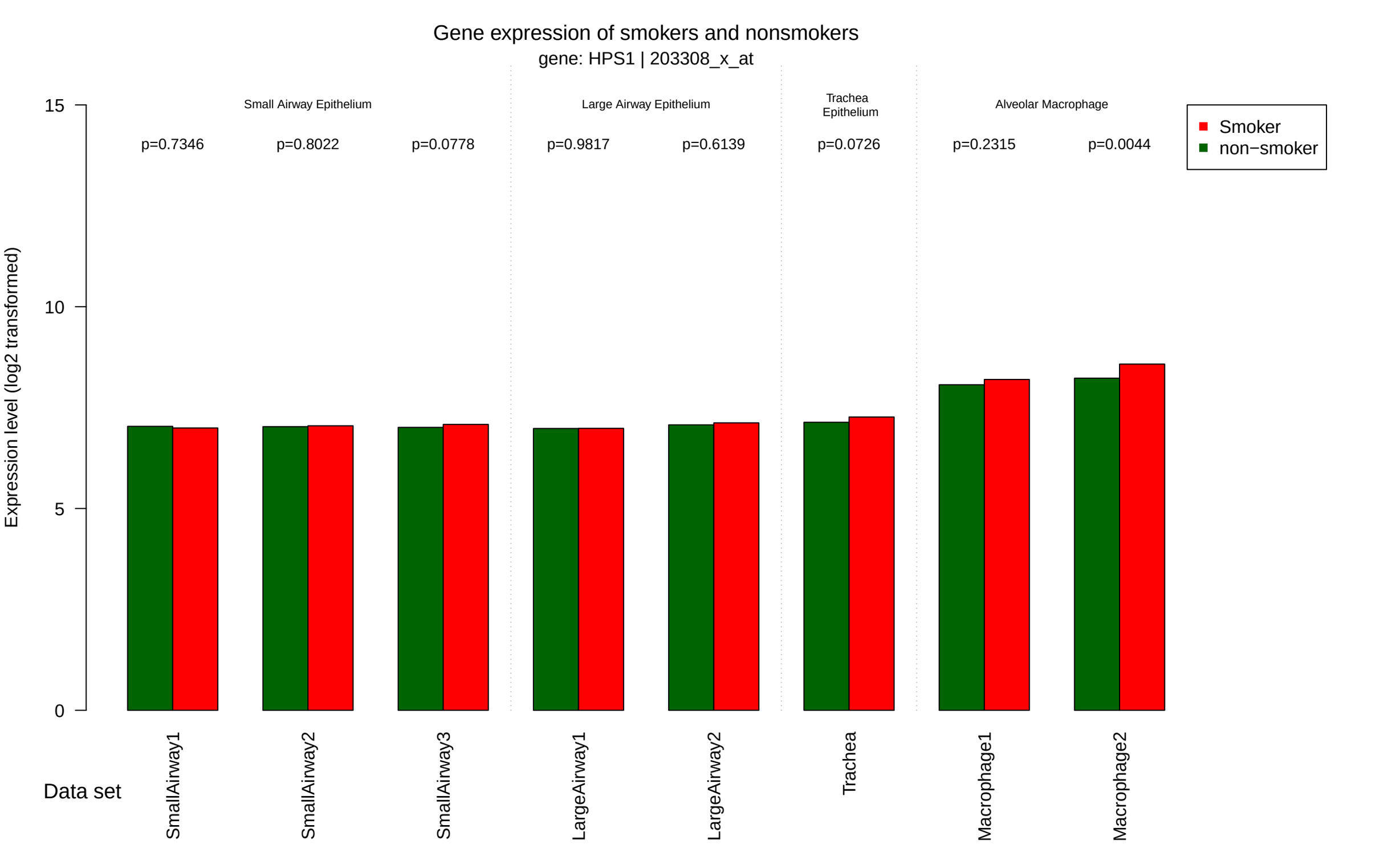
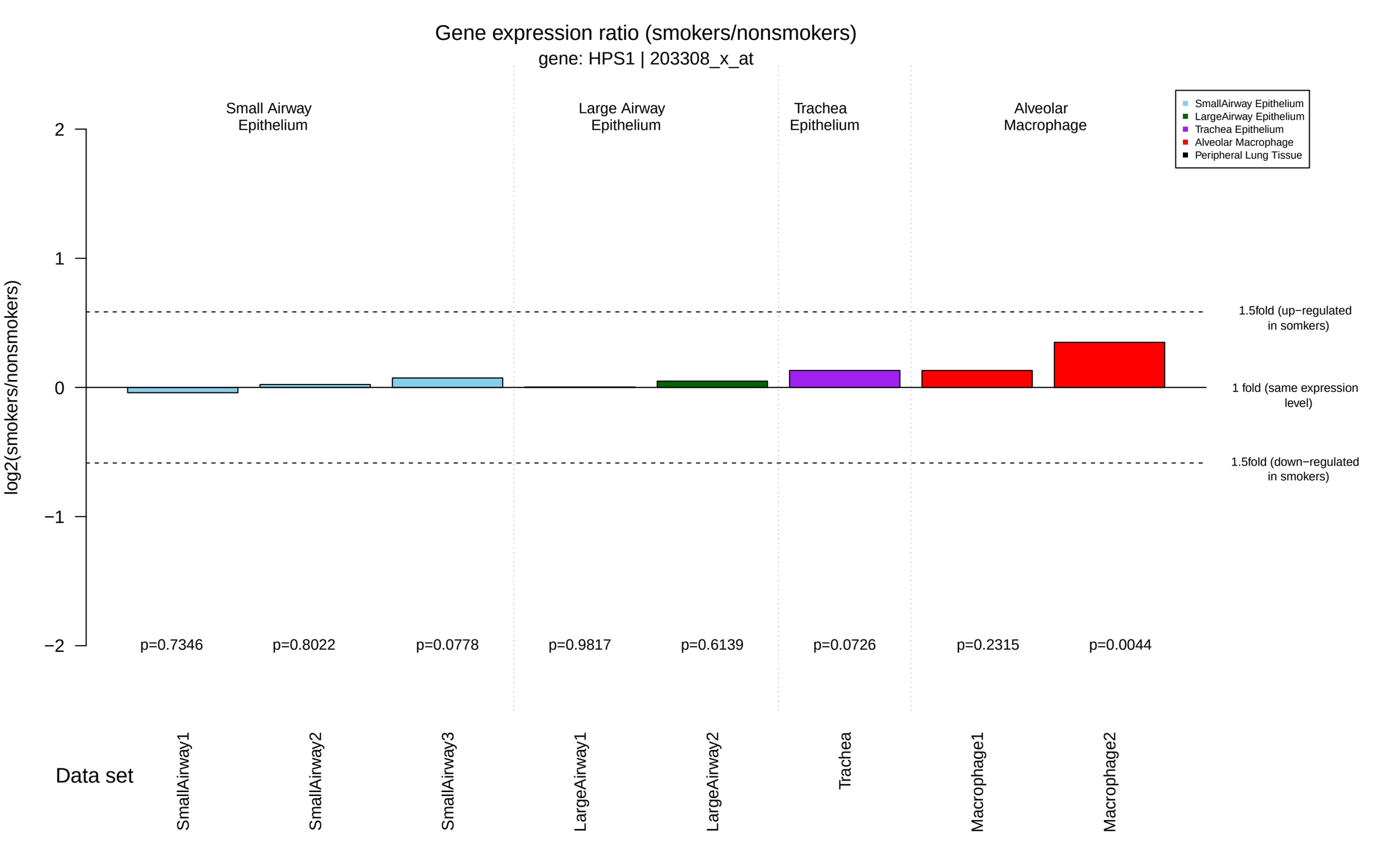
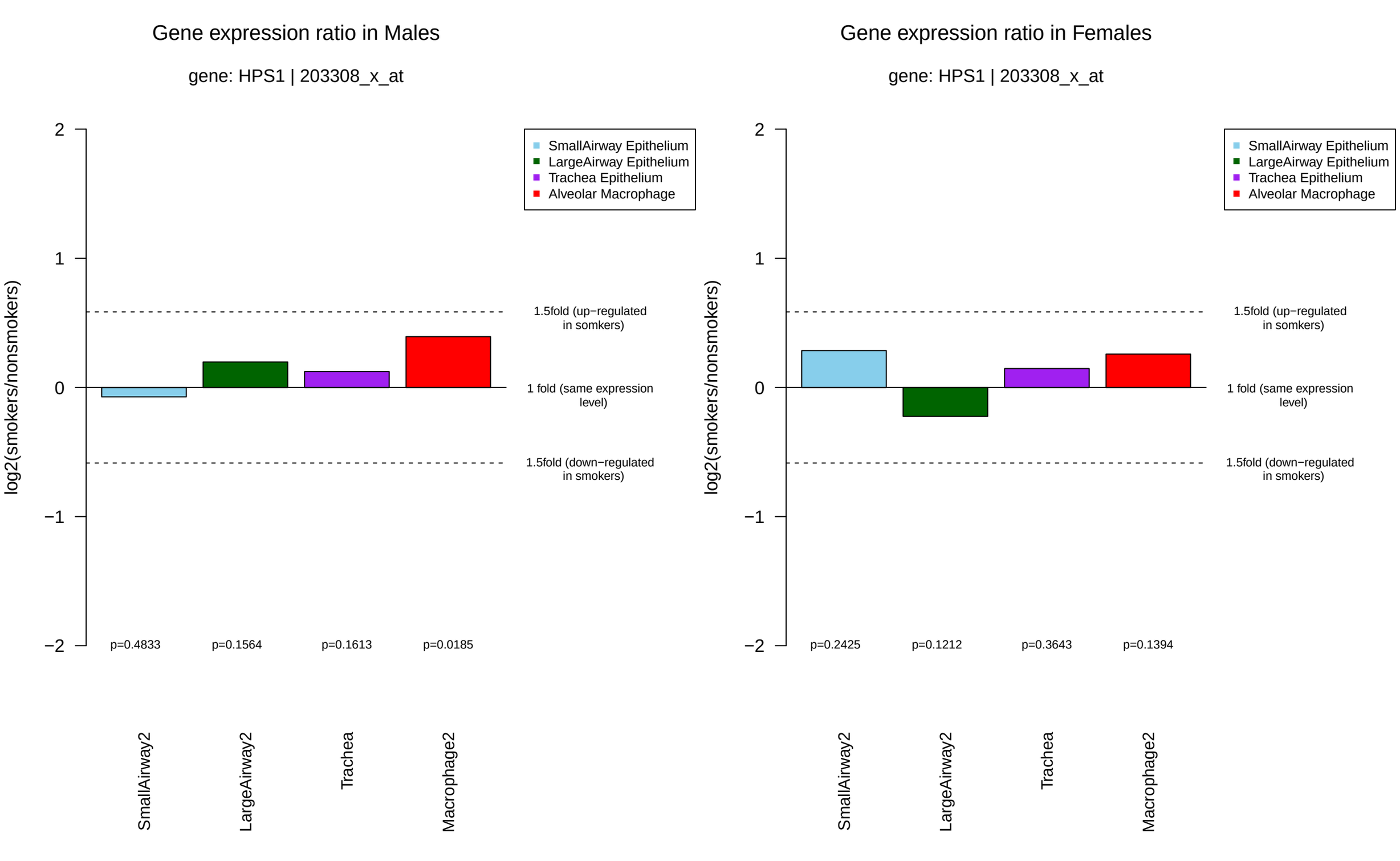
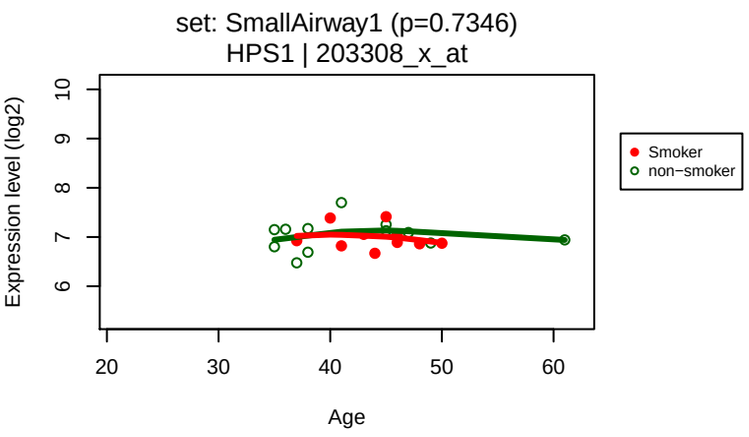
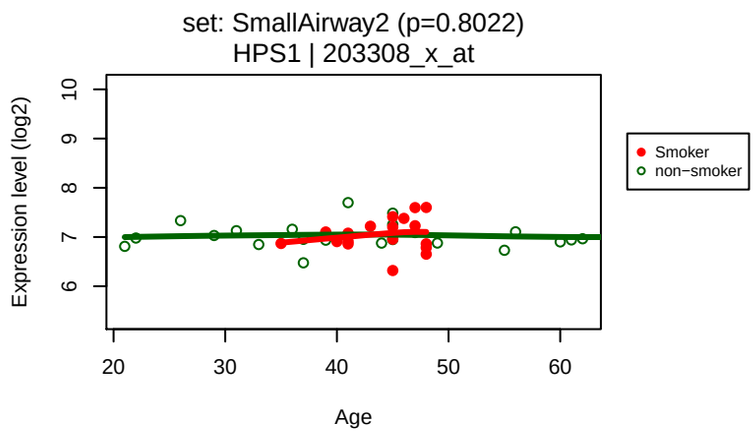
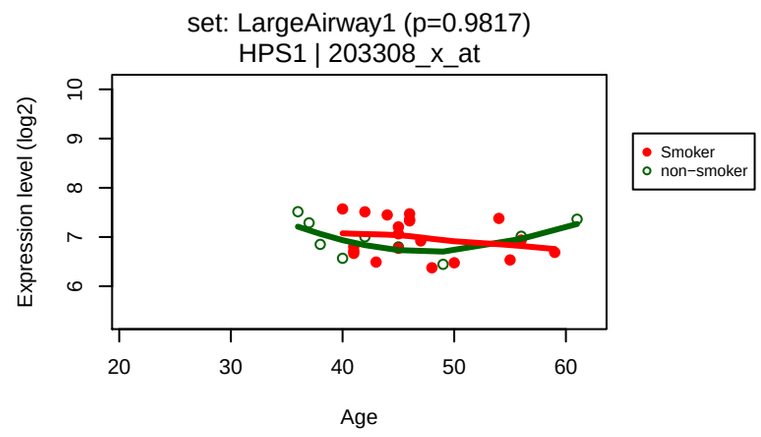

Gene expression of smokers and nonsmokers
gene: HPS1 | 203308_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7346
p=0.8022
p=0.0778
p=0.9817
p=0.6139
p=0.0726
p=0.2315
p=0.0044
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

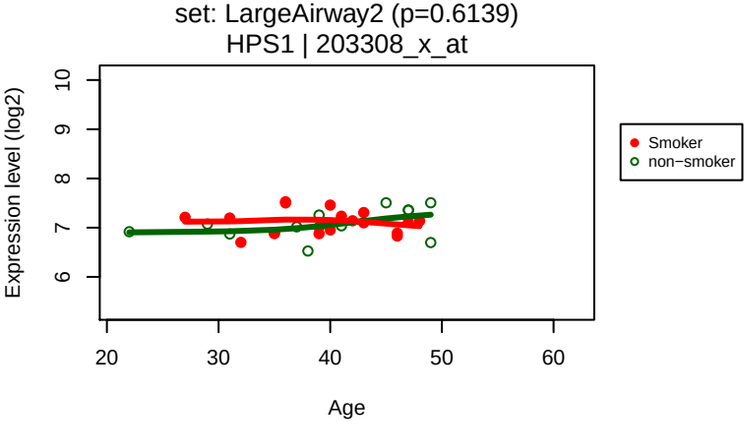

Gene expression ratio (smokers/nonsmokers)
gene: HPS1 | 203308_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7346
p=0.8022
p=0.0778
p=0.9817
p=0.6139
p=0.0726
p=0.2315
p=0.0044
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

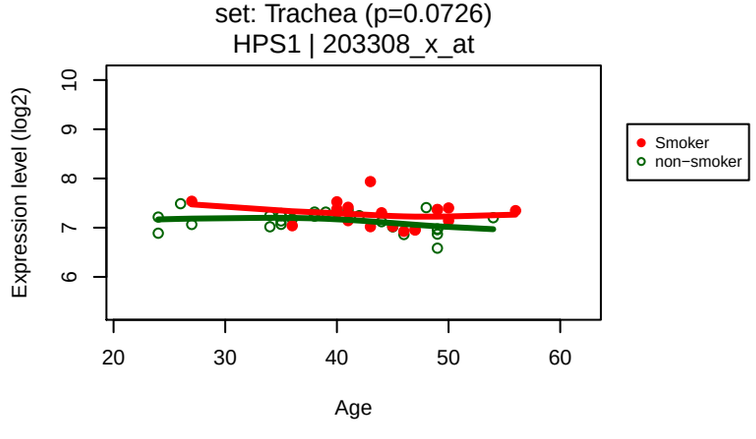

Gene expression ratio in Males
gene: HPS1 | 203308_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4833
p=0.1564
p=0.1613
p=0.0185
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HPS1 | 203308_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2425
p=0.1212
p=0.3643
p=0.1394
SmallAirway2
LargeAirway2
Trachea
Macrophage2

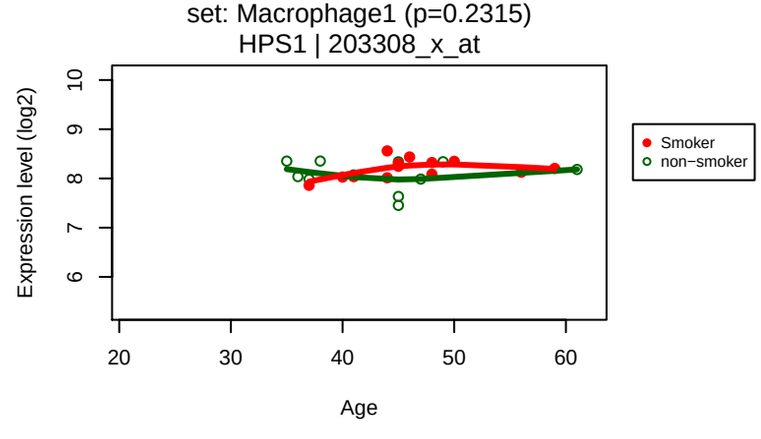

set: SmallAirway1 (p=0.7346)
HPS1 | 203308_x_at
Expression level (log2)
Age
Smoker
non−smoker

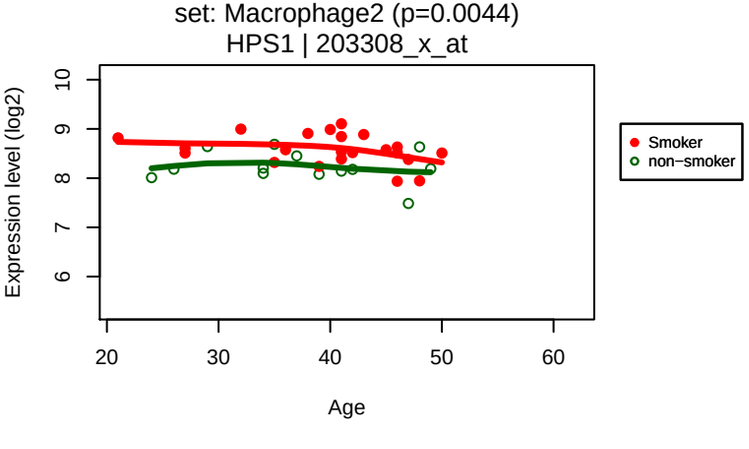

set: SmallAirway2 (p=0.8022)
HPS1 | 203308_x_at
Expression level (log2)
Age
Smoker
non−smoker

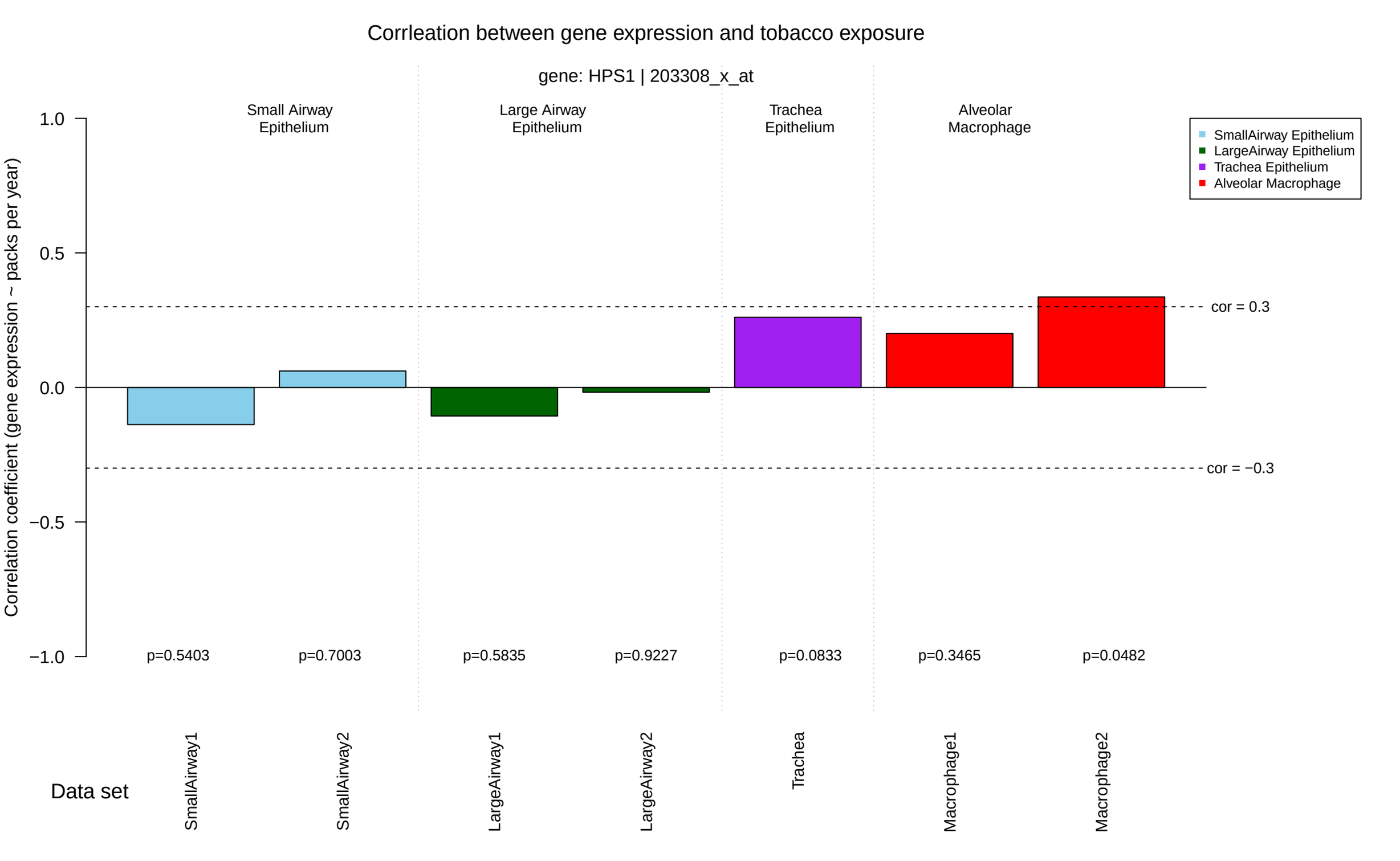

set: LargeAirway1 (p=0.9817)
HPS1 | 203308_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6139)
HPS1 | 203308_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0726)
HPS1 | 203308_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2315)
HPS1 | 203308_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0044)
HPS1 | 203308_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HPS1 | 203308_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5403
p=0.7003
p=0.5835
p=0.9227
p=0.0833
p=0.3465
p=0.0482
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2